# Nguyen Manh Hiep

Software Engineer

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## SUMMARY

I am actively seeking a Software Engineering position. Equipped with a solid technical foundation, systems thinking, and business analysis skills, I aim to bridge the gap between business needs and technological solutions. I am highly motivated to contribute my problem-solving skills to build scalable, reliable systems and deliver tangible value to the organization.

## PROJECTS

**Notarex - Notary Document OCR & Digitization** 2026 - 2026

**Software Engineer (Full-stack + AI Orchestration + Infrastructure):**

- **Description:** A platform that digitizes and manages notarial documents, allowing notary offices to extract data from scanned papers with AI OCR, edit the results online, and automatically generate finished Word documents.
- **Technologies & Tools:** Next.js 14, TypeScript, Ant Design, Zustand, TanStack Query, FastAPI (Python 3.11), SQLAlchemy 2.0 async, PostgreSQL 16, Celery, Redis, MinIO, OnlyOffice, docxtpl, Nginx, Docker, GitHub Actions.
- **Key Responsibilities & Contributions:**
  - **End-to-end Feature Development:** Built configurable form templates, an AI-powered OCR extraction flow, and an edit–save–download document workflow integrated with OnlyOffice secured by JWT, along with automatic DOCX generation using docxtpl.
  - **Authorization Design:** Implemented a 3-role permission model (admin / director / employee) scoped per notary office to keep data isolated and access clearly controlled.
  - **Background Processing:** Designed a Celery + Redis job queue using a ZSET pending queue and an atomic active-counter, with heartbeat reconciliation to detect and recover zombie jobs, ensuring OCR tasks run reliably at scale.
  - **Infrastructure & DevOps:** Managed file storage on MinIO (S3-compatible), containerized the system with Docker behind Nginx, and set up CI/CD pipelines with GitHub Actions.

**EasySale - Sales Lead Management & Automation** 2026 - 2026

**Software Engineer:**

- **Description:** A sales automation system that helps teams collect potential customers and take care of them automatically, reducing the manual effort of finding and following up with leads.
- **Technologies & Tools:** FastAPI, PostgreSQL, Chrome Extension, Zalo OA API, SendGrid, Docker, Nginx.
- **Key Responsibilities & Contributions:**
  - **Lead Collection:** Developed a Chrome Extension that scrapes business leads directly from Google Maps and pushes them into the system.
  - **Customer-care Automation:** Integrated the Zalo OA API to send automated care messages and SendGrid for email campaigns, keeping leads engaged without manual work.
  - **Portal Development:** Built separate admin and customer portals behind a unified login, and deployed the whole system with Docker and Nginx.

**AKF Farm - Agricultural Production Management (Web Admin + Mobile PWA)** 2026 - 2026

**Software Engineer:**

- **Description:** A production management platform for agriculture, combining a web admin dashboard with a mobile PWA for field workers, designed to work reliably even in low-connectivity farm environments.
- **Technologies & Tools:** React 18, Vite, TypeScript, Tailwind v4, Leaflet (+ Esri), Frappe v15, MariaDB, Redis, scheduler & queue workers, Docker Compose.
- **Key Responsibilities & Contributions:**
  - **Backend Engine:** Built a headless Frappe v15 backend (akf_farm app) with an event-driven task-generation engine based on step, frequency, and prerequisites; a 3-color “worst-of” status model; load-balanced team-leader assignment; a REST API; and 2-role authentication - verified by 132 passing tests.

- **Anti-fraud & Offline Reliability:** Implemented completion-photo verification with an in-app camera, GPS, and watermarking, plus an idempotent offline queue so field data is never lost or duplicated.
- **Field-facing Features:** Delivered a satellite heatmap (Leaflet + Esri), work calendar, notifications, and step-by-step SOP guides for workers.

## EDUCATION

**HUST - Hanoi University of Science and Technology** 2021 - 2026

Computer Science

## TECHNICAL SKILLS

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|---|---|
| **Programming** | TypeScript/JavaScript, C/C++, Java, Python. |
| **Frameworks** | React, Next.js, NestJS/ExpressJS, Spring Boot, FastAPI, Node.js. |
| **Databases** | PostgreSQL, MySQL, MariaDB, RESTful API, Celery, Redis, MinIO, Frappe. |
| **Infra & Integration** | Docker, Nginx, CI/CD (GitHub Actions), OnlyOffice, Zalo OA, SendGrid, Chrome Extension. |
| **System Design & Tech** | Systems thinking, software architecture, SOLID principles, OOP, Design Patterns, and error handling. |
| **Business Analysis** | Requirements gathering, stakeholder communication, SRS documentation, and process modeling (Use Case, Activity Diagram, ERD). |
| **Soft Skills** | Critical thinking, technical research, cross-functional teamwork, and quick adaptability to new technologies. |